

Growing Random Graphs with Quantum Rules

A classical simulation study, reproducing Jnane, Di Molfetta & Miatto (2020)

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Outline

1. The Problem

2. Model

3. Results

4. Conclusions

The Problem

How Should a Network Grow?

Classical growth rules

Models such as preferential attachment (Barabási-Albert) grow a network by a deterministic, degree-based rule: “the rich get richer.” A node’s chance of gaining a new edge depends only on its current degree.

A quantum alternative

What if, instead, a **quantum walker** explores the graph and a **measurement** decides where the new node attaches? Interference between paths has no classical, degree-based analogue.

Question

If the attachment rule is replaced by a random quantum-walk measurement, what kinds of graphs emerge – and do they reproduce theoretical predictions?

Model

A Continuous-Time Quantum Walk

$$U(t) = \exp(-iAt), \quad (1)$$

the adjacency matrix A acts as the Hamiltonian.

$$P_t(v) = |\langle v | U(t) | \psi_0 \rangle|^2, \quad (2)$$

the Born rule gives the probability of measuring the walker at node v .

$$t \sim \text{Exp}(\tau), \quad (3)$$

the collapse time itself is random.

- Small τ : the walker is measured almost immediately $\rightarrow$ the graph tends to collapse into a **star**.
- Large τ : the walker spreads out before measurement $\rightarrow$ new edges land far

The Growth Algorithm

Start from a graph with a single node;

for each new node to add **do**

 Draw a random collapse time $t \sim \text{Exp}(\tau)$;

 Evolve the walker: $\psi(t) = U(t) \psi_0$;

 Measure with the Born rule $\rightarrow$ node v ;

 Attach the new node to v ;

return the grown graph;

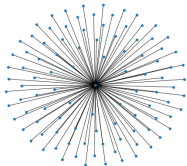
With several walkers, the evolve-and-measure step runs once per walker before the new node is added, so more than one edge can form at once.

Results

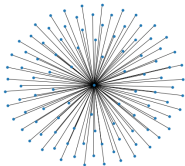
Sample Graphs Across τ

1 walker — 100 steps — growth into trees

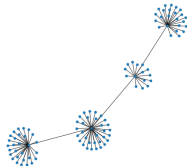
$\tau = 0.001$



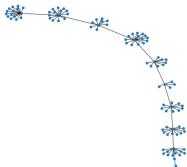
$\tau = 0.01$



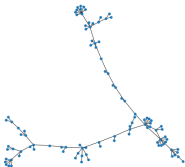
$\tau = 0.05$



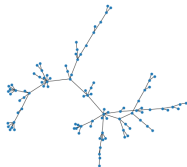
$\tau = 0.1$



$\tau = 0.5$



$\tau = 10$



1

The Problem

2

Model

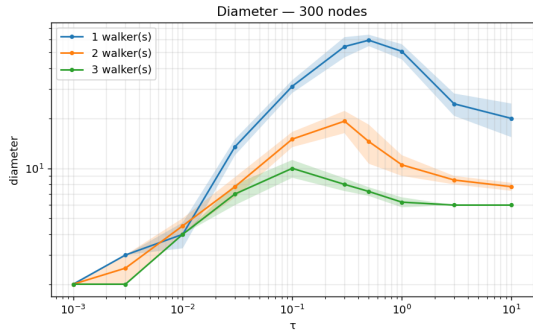
3

Results

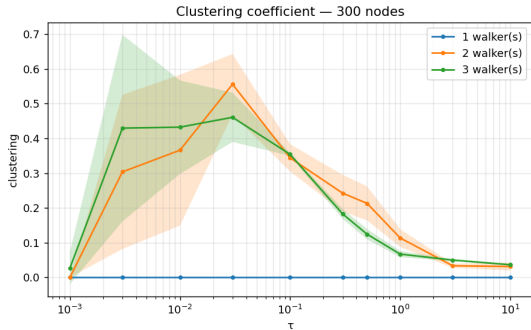
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Conclusions

Diameter and Clustering vs. τ

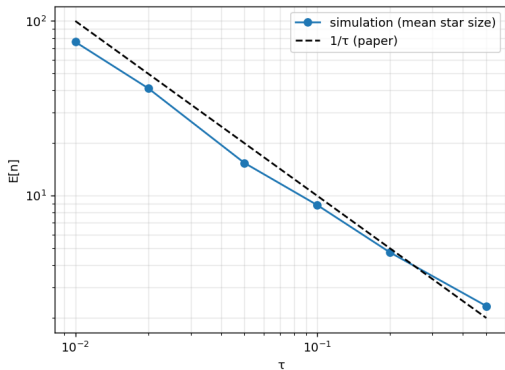


Diameter peaks at intermediate τ ($\approx 0.1-1$); more walkers shorten it by adding shortcuts.



Clustering is essentially zero for one walker, and decreases with τ for two or three walkers.

Checking $E[n] \approx 1/\tau$



τ	$E[n]$	$1/\tau$	ratio
0.01	75.9	100.0	0.76
0.02	41.3	50.0	0.83
0.05	15.4	20.0	0.77
0.1	8.9	10.0	0.89
0.2	4.7	5.0	0.95
0.5	2.3	2.0	1.17

Table: Simulated mean star size vs. the $1/\tau$ prediction.

The ratio $E[n] \cdot \tau$ stays close to a constant for small τ , confirming the $1/\tau$ scaling; it drifts upward near $\tau = 0.5$, where the small- τ approximation starts to break down.

Conclusions

Key Findings

Claim from the paper	Observed here
Small $\tau \rightarrow$ large stars	Reproduced
Mean star size $E[n] \approx 1/\tau$	Reproduced (factor ≈ 0.8 - 0.9)
Diameter maximal for $0.1 < \tau < 1$	Reproduced
Clustering is 0 for one walker, decreases with τ	Reproduced
Degree distribution $\rightarrow$ power law for large τ	Partial: shape matches, no formal fit

In short: unlike degree-based preferential attachment, the quantum rule ties graph structure to a single physical parameter, τ , through interference and measurement – not through vertex degree.

Summary and References

What this is. A classical simulation of a quantum-walk growth rule (dense/sparse matrix exponentials, Born-rule sampling) – no quantum hardware involved.

Open questions. A formal power-law fit for the degree distribution; larger graphs; a possible phase transition near $\tau \approx 1$.

Reference.

Jnane, H.; Di Molfetta, G.; Miatto, F. M. (2020). Growing Random Graphs with Quantum Rules. EPTCS 315, pp. 38–47. DOI: 10.4204/EPTCS.315.4

Thank you.

That's all folks!

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